Technical Report: Synthetic FFR Benchmark Generation

1. Overview

This document details the mathematical and physiological principles used to generate the synthetic FFR datasets for the FFR-GAN project. The goal was to create a controlled environment to test the ability of a Conditional GAN to correct systematic biases in hemodynamic modeling.

2. Anatomical Parameter Space

We simulated a cohort of 400 virtual patients. The anatomical parameters were sampled from distributions observed in clinical CAD populations (e.g., ASPIRE database):

Parameter	Range	Distribution
Vessel Angle	30° - 90°	Discrete (30, 45, 60, 90)
Stenosis Severity	20% - 90%	Uniform Random
Stenosis ID	1 - 100	Categorical Identifier

3. Hemodynamic Modeling (The Physics Engine)

The FFR values were calculated using a modified Poiseuille-derived pressure drop equation.

3.1 Reference Standard (Diameter-based)

The “Real” data assumes a high-fidelity measurement where the pressure drop (ΔP) is highly sensitive to the narrowest diameter (D): $\Delta P_{ref} = C_{lesion} \cdot (S^{3.5}) + C_{vessel} \cdot (\frac{\theta}{90})$ $FFR_{ref} = 1.0 - \Delta P_{ref} + \epsilon * Where S is severity, \theta is angle, and \epsilon$ $is physiological noise$(\sigma=0.02$).*

3.2 Biased Model (Area-based)

The “Fake” data simulates a simplified area-based assumption which underestimates the resistance of the lesion: $\Delta P_{bias} = C_{low} \cdot (S^{2.0}) + C_{min} \cdot (\frac{\theta}{90})$ This results in the characteristic “High FFR Bias” (Mean ~0.93) often seen in lower-fidelity imaging modalities.

4. Physiological Constraints

To maintain clinical realism, the following safeguards were implemented in the generation script (`generate_ffr_benchmarks.py`):

- **FFR Floor**: All values are clipped at **0.40**, representing the lower limit of biological viability.
- **Monotonicity**: The engine ensures that increasing stenosis severity always results in a non-increasing FFR value.

5. Conclusion

By generating both datasets from a single “Virtual Patient” geometry, we created a perfect ground-truth mapping. This allows the cGAN to learn the specific transformation required to move from a biased area-based space to a realistic diameter-based physiological space.